

SECTION 5.3: THE FUNDAMENTAL THEOREM OF CALCULUS (DAY 2ISH)
EVEN MORE PRACTICE!

1. Find the derivative of each function below.

(a) $g(x) = \int_{-1}^x t^2 e^t dt$

(d) $l(x) = \int_{\sqrt{x}}^1 \tan(t) dt$

(b) $h(x) = \int_0^{x^2+1} \sin(t) dt$

(e) $m(\theta) = \int_0^{\tan \theta} \sec^2 y dy$

(c) $k(x) = \int_1^{\sqrt{x}} \tan(t) dt$

(f) $n(x) = \int_{-1}^x \frac{t^2}{t^2+4} dt - \int_3^{x^2} \frac{t^2}{t^2+4} dt$

2. Evaluate each definite integral below using the Fundamental Theorem of the Calculus, Part 2.

- Hint 1: *You may need to do some rewriting to get it into the form of something you know how to take the antiderivative of.*
- Hint 2: *How does the derivative of $f(5t)$ relate to the derivative of $f(t)$?*

(a) $\int_0^2 x(x-3) dx$

(d) $\int_1^2 \left(\frac{1}{x} - e^{-x} \right) dx$

(b) $\int_{\pi/2}^0 \frac{1 + \cos(2t)}{2} dt$

(e) $\int_2^4 x^{\pi-1} dx$

(c) $\int_{-\sqrt{3}}^{\sqrt{3}} (t+1)(t^2+4) dt$

(f) $\int_{-1}^0 \pi^{x-1} dx$